

Requirements Traceability Matrix

ADLink Multi-Channel Advertising Management Platform

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1 Introduction

This Requirements Traceability Matrix (RTM) establishes comprehensive traceability between requirements specified in the Software Requirements Specification (SRS) and their corresponding design elements, diagrams, use cases, test cases, and implementation components for the ADLink Multi-Channel Advertising Management Platform.

1.1 Purpose

The RTM serves to:

- Ensure all requirements are addressed in design and implementation
- Facilitate impact analysis for requirement changes
- Support verification and validation activities
- Maintain bidirectional traceability throughout the SDLC
- Enable progress tracking and requirement coverage analysis

1.2 Document Conventions

- **Req ID:** Unique requirement identifier from SRS
- **Priority:** High (H), Medium (M), Low (L)
- **Status:** Approved (A), In Progress (IP), Completed (C), Tested (T)
- **UC:** Use Case
- **TC:** Test Case
- **DFD:** Data Flow Diagram

2 Functional Requirements Traceability

2.1 User Management Requirements

Req ID	Description	Use Case	Design Artifact	Diagram Reference	Test Case	Pri
REQ-F-1.1	Email/password authentication	UC-01, UC-02	AuthController, UserService, Fire-base Auth	Use Case Diagram, Sequence Diagram (Auth), Class Diagram (User)	TC-001, TC-002, TC-003	H
REQ-F-1.2	Role selection: Business Owner, Service Provider, Asset Owner, Admin	UC-01, UC-03	UserRoleService, RoleEnum, Profile-Controller	Use Case Diagram, Class Diagram (User Hierarchy), Activity Diagram	TC-004, TC-005, TC-006	H
REQ-F-1.3	User sub-classification by role type	UC-03	UserSubTypeService, UserClassification Model	Class Diagram, DFD Level 1 (User Mgmt)	TC-007, TC-008	M
REQ-F-1.4	Role-based access control	All UCs	RBACMiddleware, PermissionService, AuthGuard	Component Diagram, Activity Diagram, Sequence Diagram	TC-009, TC-010, TC-011	H
REQ-F-2.1	Profile CRUD operations	UC-03, UC-04	ProfileController, ProfileService, ProfileModel	Use Case Diagram, Sequence Diagram (Profile), Class Diagram	TC-012, TC-013, TC-014	H
REQ-F-2.2	Role-specific profile fields	UC-03	DynamicFormGenerator, ProfileFieldService, Role-based Profile Models	Class Diagram (Profile inheritance), DFD Level 2	TC-015, TC-016, TC-017	M

Table 1: User Management Requirements Traceability

2.2 Asset and Service Management

Req ID	Description	Use Case	Design Artifact	Diagram Reference	Test Case	Pri
REQ-F-3.1	Asset listing CRUD	UC-05, UC-06	AssetController, AssetService, Asset-Model	Use Case Diagram, DFD Level 1 (Billboard Mgmt), Class Diagram	TC-018, TC-019, TC-020	H
REQ-F-3.2	Asset details: location, pricing, availability, media	UC-05	AssetModel, MediaUploadService, LocationService, PricingModel	Class Diagram, DFD Level 2, Sequence Diagram (Asset Creation)	TC-021, TC-022, TC-023	H
REQ-F-3.3	Service listing management	UC-07	ServiceController, ServiceModel, ServiceManager	Use Case Diagram, DFD Level 1, Component Diagram	TC-024, TC-025, TC-026	H

Table 2: Asset and Service Management Traceability

2.3 Service Discovery

Req ID	Description	Use Case	Design Artifact	Diagram Reference	Test Case	Pri
REQ-F-4.1	Browse services and assets	UC-08, UC-09	SearchController, BrowseService, ServiceRepository	Use Case Diagram, Activity Diagram (Discovery), DFD Level 1	TC-027, TC-028	H
REQ-F-4.2	Search/filter by type, location, budget, ratings	UC-09	FilterService, SearchEngine, QueryBuilder	Activity Diagram, DFD Level 1, Component Diagram	TC-029, TC-030, TC-031	H
REQ-F-4.3	View detailed provider profiles	UC-10	ProfileViewController, ProviderDetailService	Sequence Diagram (Profile View), Use Case Diagram	TC-032, TC-033	M

Table 3: Service Discovery Traceability

2.4 Booking and Campaign Management

Req ID	Description	Use Case	Design Artifact	Diagram Reference	Test Case	Pri
REQ-F-5.1	Digital booking of advertising services	UC-11, UC-12, UC-13	BookingController, ReservationService, BookingModel	Use Case Diagram, Sequence Diagram (Booking), DFD Level 1	TC-034, TC-035, TC-036	H
REQ-F-5.2	Campaign config: duration, customization, pricing	UC-11	CampaignConfigService, PricingCalculator, BookingCustomizer	Sequence Diagram, Activity Diagram (Booking), Class Diagram	TC-037, TC-038	H
REQ-F-5.3	Record booking details	UC-11, UC-14	BookingRepository, TransactionLogger, BookingHistory	DFD Level 2, Class Diagram (Booking)	TC-039, TC-040	M
REQ-F-6.1	Campaign dashboard for business owners	UC-15, UC-16	CampaignController, DashboardService, CampaignModel	Use Case Diagram, DFD Level 1 (Campaign), Component Diagram	TC-041, TC-042, TC-043	H
REQ-F-6.2	Provider booking management	UC-17	ProviderDashboard, BookingManager, NotificationService	Use Case Diagram, Sequence Diagram, Class Diagram	TC-044, TC-045	M

Table 4: Booking and Campaign Management Traceability

2.5 Payment Processing

Req ID	Description	Use Case	Design Artifact	Diagram Reference	Test Case	Pri
REQ-F-7.1	Secure in-app payment	UC-18	PaymentController, PaymentGatewayService, TransactionProcessor	Use Case Diagram, Sequence Diagram (Payment), Context Diagram	TC-046, TC-047, TC-048	H
REQ-F-7.2	Transaction history and invoices	UC-19	TransactionHistoryService, InvoiceGenerator, PaymentRepository	DFD Level 1, Class Diagram (Payment), Component Diagram	TC-049, TC-050	M
REQ-F-7.3	Link payments to campaigns	UC-18	PaymentBookingLinker, CampaignPaymentService	Class Diagram, DFD Level 2	TC-051, TC-052	H

Table 5: Payment Processing Traceability

2.6 Reviews and Administration

Req ID	Description	Use Case	Design Artifact	Diagram Reference	Test Case	Pri
REQ-F-8.1	Submit ratings and reviews	UC-20	ReviewController, RatingService, ReviewModel	Use Case Diagram, DFD Level 1, Class Diagram (Review)	TC-053, TC-054	M
REQ-F-8.2	Display ratings on profiles	UC-10	ReviewDisplayService, RatingAggregator	Sequence Diagram, Component Diagram	TC-055, TC-056	M
REQ-F-9.1	Admin: user mgmt, monitoring, disputes	UC-21, UC-22, UC-23	AdminController, UserManager, SystemMonitor, DisputeResolver	Use Case Diagram, Context Diagram, Component Diagram	TC-057, TC-058, TC-059	H
REQ-F-10.1	Support requirement updates	UC-24	ConfigurationManager, SystemUpdater	Activity Diagram, DFD Level 0	TC-060	L

Table 6: Reviews and Administration Traceability

3 Non-Functional Requirements Traceability

3.1 Performance and Security

Req ID	Description	Design Strategy	Implementation	Test Case	Pri
REQ-NF-1.1	95% actions respond within 1 sec	Caching, Async Processing, Indexing	Redis Cache, Query Optimization, CDN	TC-061, TC-062, TC-063	H
REQ-NF-1.2	Handle 1000+ concurrent users	Load Balancing, Horizontal Scaling	Firebase Auto-scaling, Efficient APIs, Rate Limiting	TC-064, TC-065	H
REQ-NF-1.3	Home screen loads in 3 sec	Lazy Loading, Code Splitting	Progressive Loading, Image Compression, Minification	TC-066, TC-067	M
REQ-NF-2.1	Secure authentication and storage	Encryption, Secure Protocols	Firebase Auth, HTTPS/TLS, Encrypted DB	TC-068, TC-069, TC-070	H
REQ-NF-2.2	Password complexity (8+ chars)	Password Validation, Strength Meter	Regex validation, bcrypt hashing, Salt	TC-071, TC-072	H
REQ-NF-2.4	RBAC implementation	Authorization Framework, Permission Matrix	RBAC Middleware, JWT tokens, Role hierarchy	TC-073, TC-074, TC-075	H

Table 7: Performance and Security Requirements Traceability

3.2 Reliability and Availability

Req ID	Description	Design Strategy	Implementation	Test Case	Pri
REQ-NF-3.1	Consistent performance	Error Handling, Graceful Degradation	Try-catch, Fallback mechanisms, Logging	TC-076, TC-077	H
REQ-NF-3.2	Minimize critical failures	Transaction Management, Atomicity	DB transactions, Rollback, Validation	TC-078, TC-079	H
REQ-NF-3.3	MTTR ≤ 2 hours	Monitoring, Alerting, Auto-recovery	Health checks, Automated alerts, Backups	TC-080, TC-081	H
REQ-NF-3.4	99.9% success rate for critical ops	Redundancy, Retry Logic	Payment retry, Confirmation system, Audit logs	TC-082, TC-083, TC-084	H
REQ-NF-5.1	99.5% uptime, max 4hr maintenance	High Availability Architecture	Firebase SLA, Redundant systems, Maintenance windows	TC-091, TC-092	H
REQ-NF-5.2	Reliable booking/campaign mgmt	System Stability, Data Persistence	Reliable DB, Transaction integrity, Backups	TC-093, TC-094	H

Table 8: Reliability and Availability Requirements Traceability

3.3 Usability and Compatibility

Req ID	Description	Design Strategy	Implementation	Test Case	Pri
REQ-NF-4.1	30min training, SUS ≥ 70	Intuitive UI/UX, Onboarding	Tutorial screens, Help tooltips, Clear navigation	TC-085, TC-086	H
REQ-NF-4.2	Accessible to non-technical users	Simplified Interface, Plain Language	Visual guides, Context help, Minimal jargon	TC-087, TC-088	H
REQ-NF-4.3	Clean, intuitive interface	UI Design Principles, Consistency	Material Design, Consistent theme, Responsive	TC-089, TC-090	M
REQ-NF-7.1	Android and iOS support	Cross-platform Framework	Flutter, Platform-specific APIs, Responsive design	TC-099, TC-100	H
REQ-NF-7.2	Consistent across devices	Platform Abstraction, Testing	Cross-platform testing, Device emulators	TC-101, TC-102	H
REQ-NF-7.3	Support 720x1280 to 1440x3040px	Responsive Design, Flexible Layouts	Adaptive UI, Media queries, Scalable assets	TC-103, TC-104	M

Table 9: Usability and Compatibility Requirements Traceability

3.4 Scalability

Req ID	Description	Design Strategy	Implementation	Test Case	Pri
REQ-NF-6.1	Scale users/services across Pakistan	Modular Architecture, DB Scalability	Microservices, NoSQL, Cloud infrastructure	TC-095, TC-096	H
REQ-NF-6.2	Geographic expansion without changes	Flexible Data Model, Location-agnostic	Configurable regions, Multi-tenancy, Location services	TC-097, TC-098	M

Table 10: Scalability Requirements Traceability

4 Requirements Coverage Summary

4.1 Functional Requirements Coverage

Requirement Category	Total	Traced	Coverage %
User Management	6	6	100%
Asset & Service Management	3	3	100%
Service Discovery	3	3	100%
Booking & Campaign	5	5	100%
Payment Processing	3	3	100%
Reviews	2	2	100%
Administration	2	2	100%
Total Functional	24	24	100%

Table 11: Functional Requirements Coverage

4.2 Non-Functional Requirements Coverage

Requirement Category	Total	Traced	Coverage %
Performance	3	3	100%
Security	3	3	100%
Reliability	4	4	100%
Usability	3	3	100%
Availability	2	2	100%
Scalability	2	2	100%
Compatibility	3	3	100%
Total Non-Functional	20	20	100%

Table 12: Non-Functional Requirements Coverage

4.3 Overall Traceability Statistics

Metric	Value
Total Requirements	44
Requirements Traced to Design	44
Requirements Traced to Test Cases	44
Requirements Traced to Diagrams	44
Overall Traceability Coverage	100%

Table 13: Overall Traceability Statistics

5 Diagram-to-Requirement Mapping

5.1 Use Case Diagram Mapping

Use Case	Related Requirements
UC-01: Register Account	REQ-F-1.1, REQ-F-1.2
UC-02: Login	REQ-F-1.1, REQ-F-1.4
UC-03: Setup Profile	REQ-F-1.2, REQ-F-1.3, REQ-F-2.1, REQ-F-2.2
UC-04: Manage Profile	REQ-F-2.1
UC-05: List Asset	REQ-F-3.1, REQ-F-3.2
UC-06: Manage Assets	REQ-F-3.1
UC-07: Manage Services	REQ-F-3.3
UC-08: Browse Services	REQ-F-4.1
UC-09: Search Services	REQ-F-4.1, REQ-F-4.2
UC-10: View Provider Details	REQ-F-4.3, REQ-F-8.2
UC-11: Book Service	REQ-F-5.1, REQ-F-5.2, REQ-F-5.3
UC-12: Book Billboard	REQ-F-5.1
UC-13: Book Vehicle Ad	REQ-F-5.1
UC-14: View Bookings	REQ-F-5.3
UC-15: Manage Campaigns	REQ-F-6.1
UC-16: View Campaign Status	REQ-F-6.1
UC-17: View Provider Bookings	REQ-F-6.2
UC-18: Process Payment	REQ-F-7.1, REQ-F-7.3
UC-19: View Transaction History	REQ-F-7.2
UC-20: Submit Review	REQ-F-8.1
UC-21: Manage Users	REQ-F-9.1
UC-22: Monitor System	REQ-F-9.1
UC-23: Resolve Disputes	REQ-F-9.1
UC-24: Update Requirements	REQ-F-10.1

Table 14: Use Case to Requirements Mapping

5.2 Diagram Coverage Analysis

Diagram Type	Requirements Covered
Use Case Diagram	24 Functional Requirements
Class Diagram	24 Functional Requirements
Sequence Diagrams	18 Functional Requirements
Activity Diagrams	15 Functional Requirements
Data Flow Diagrams (DFD 0, 1, 2)	24 Functional Requirements
Component Diagram	20 Functional + 12 Non-Functional
Context Diagram	All Requirements (System Boundary)

Table 15: Diagram Coverage Analysis

6 Change Impact Analysis Matrix

6.1 Requirement Dependencies

Req ID	Depends On	Impacts	Risk Level
REQ-F-1.1	None (Foundation)	REQ-F-1.2, REQ-F-1.4, REQ-F-2.1, All authenticated features	High
REQ-F-1.4	REQ-F-1.1, REQ-F-1.2	All functional requirements (access control)	High
REQ-F-3.1	REQ-F-1.1, REQ-F-1.2, REQ-F-2.1	REQ-F-4.1, REQ-F-4.2, REQ-F-5.1	High
REQ-F-5.1	REQ-F-1.1, REQ-F-4.1, REQ-F-3.1	REQ-F-7.1, REQ-F-6.1	High
REQ-F-7.1	REQ-F-5.1, Payment Gateway APIs	REQ-F-7.2, REQ-F-7.3, REQ-F-6.1	Critical
REQ-NF-2.1	All Infrastructure	All Requirements	Critical
REQ-NF-1.2	System Architecture	All Performance Requirements	High

Table 16: Requirement Dependencies and Impact Analysis

7 Traceability Maintenance

7.1 Version Control

Version	Date	Author	Changes
1.0	Nov 28, 2025	M. Hamza, B. Khan	Initial RTM creation

Table 17: RTM Version History

7.2 Maintenance Guidelines

1. **Requirement Changes:** When a requirement changes, update all corresponding entries in the RTM including design artifacts, test cases, and diagrams.
2. **New Requirements:** Add new rows to the appropriate traceability table and update the coverage summary.
3. **Requirement Deletion:** Mark as "Deprecated" rather than deleting to maintain audit trail.
4. **Regular Reviews:** Conduct bi-weekly reviews during development to ensure RTM accuracy.
5. **Status Updates:** Update requirement status as development progresses (Approved → In Progress → Completed → Tested).

7.3 Traceability Verification Checklist

All requirements from SRS are included in RTM

Each requirement maps to at least one use case

Each requirement maps to at least one design artifact

Each requirement maps to at least one test case

Each requirement references appropriate diagrams

Priority levels are assigned to all requirements

Bidirectional traceability is maintained

Coverage statistics are accurate

8 Appendix A: Requirement-to-Test Case Mapping

8.1 Test Case Summary

Test Category	Test Cases	Requirements Covered
Authentication Tests	TC-001 to TC-011	REQ-F-1.1 to REQ-F-1.4
Profile Management Tests	TC-012 to TC-017	REQ-F-2.1, REQ-F-2.2
Asset/Service Tests	TC-018 to TC-026	REQ-F-3.1 to REQ-F-3.3
Discovery Tests	TC-027 to TC-033	REQ-F-4.1 to REQ-F-4.3
Booking Tests	TC-034 to TC-045	REQ-F-5.1 to REQ-F-6.2
Payment Tests	TC-046 to TC-052	REQ-F-7.1 to REQ-F-7.3
Review Tests	TC-053 to TC-056	REQ-F-8.1, REQ-F-8.2
Admin Tests	TC-057 to TC-060	REQ-F-9.1, REQ-F-10.1
Performance Tests	TC-061 to TC-067	REQ-NF-1.1 to REQ-NF-1.3
Security Tests	TC-068 to TC-075	REQ-NF-2.1, REQ-NF-2.2, REQ-NF-2.4
Reliability Tests	TC-076 to TC-084	REQ-NF-3.1 to REQ-NF-3.4
Usability Tests	TC-085 to TC-090	REQ-NF-4.1 to REQ-NF-4.3
Availability Tests	TC-091 to TC-094	REQ-NF-5.1, REQ-NF-5.2
Scalability Tests	TC-095 to TC-098	REQ-NF-6.1, REQ-NF-6.2
Compatibility Tests	TC-099 to TC-104	REQ-NF-7.1 to REQ-NF-7.3
Total	104	44 Requirements

Table 18: Test Case Coverage Summary

9 Appendix B: Design Artifact Reference

9.1 Key Design Components

- **Controllers:** AuthController, ProfileController, AssetController, ServiceController, BookingController, CampaignController, PaymentController, ReviewController, AdminController
- **Services:** UserService, RBACMiddleware, AssetService, SearchEngine, FilterService, BookingService, PaymentGatewayService, NotificationService
- **Models:** User, Profile, Asset, Service, Booking, Campaign, Payment, Review
- **Repositories:** UserRepository, AssetRepository, ServiceRepository, BookingRepository, PaymentRepository
- **Utilities:** MediaUploadService, LocationService, PricingCalculator, InvoiceGenerator, ValidationService

10 Conclusion

This Requirements Traceability Matrix provides comprehensive coverage of all functional and non-functional requirements specified in the ADLink SRS document. With 100% traceability from requirements through design artifacts, diagrams, and test cases, this RTM serves as a foundational document for verification, validation, and change management throughout the software development lifecycle.

The RTM will be maintained and updated as the project progresses, ensuring continuous alignment between requirements, design, implementation, and testing activities.